

Mathematical definition of StaBiCut stability metrics and scoring components

StaBiCut (Stable and Biologically aware Cutoff prioritization) was implemented as a deterministic, direction-aware framework for survival cutoff selection and post-selection prioritization. In the current implementation, the pipeline operates on $\log_2(\text{TPM}+1)$ -transformed expression data, removes genes with very low mean expression, restricts survival analyses to tumor samples, and performs per-gene cutoff analysis on winsorized tumor expression values to limit the influence of extreme outliers. For each gene, a per-sample analysis table $D = \{(xi, ti, \delta_i)\}_{i=1}^n$ was constructed, where xi denotes winsorized tumor expression, ti denotes follow-up time in months, and δ_i denotes the event indicator. Tumor-normal comparisons were computed from the full $\log_2(\text{TPM}+1)$ expression values in tumor and normal samples rather than the winsorized tumor-only analysis table.

An optional direction-prior layer was applied before cutpoint scanning. Let

$$\Delta_{TN} = \text{median}(x^{\text{tumor}}) - \text{median}(x^{\text{normal}}),$$

denote the tumor–normal median difference, where x^{tumor} and x^{normal} denote the full $\log_2(\text{TPM}+1)$ expression values in tumor and normal samples, respectively. Let β_{cont} denote the coefficient from the continuous Cox model

$$h(t | x) = h_0(t)\exp(\beta_{\text{cont}}x).$$

In the current code path, expected direction was determined in the following order. First, if a gene-specific prior table contained a valid expected direction, that value was used directly. In the present study, all 11 candidate genes had been pre-nominated from longitudinal transcriptomic and proteomic analyses, so gene-specific directional priors were available before survival stratification. The cutoff scan was accordingly constrained by the expected directions summarized in Direction prior summary of online supplementary table 7, with `adverse_high` assigned to candidates showing progressive upregulation and

protective_high assigned to candidates showing progressive downregulation in the CAC multi-omics profiles. More generally, when such prior information is unavailable, StaBiCut can infer expected direction only when the tumor-normal shift and the sign of the continuous Cox coefficient are concordant; otherwise, no direction constraint is imposed. Second, only if no valid gene-specific prior was available, fallback inference was allowed when Δ_{TN} and β_{cont} were both available and concordant: $\Delta_{TN} < 0$ and $\beta_{cont} < 0$ implied protective_high, whereas $\Delta_{TN} > 0$ and $\beta_{cont} > 0$ implied adverse_high. Third, a force direction branch was implemented as a stress-testing mode based on the sign of Δ_{TN} alone, but this branch was not used in the present analysis. This fallback rule was implemented in the StaBiCut workflow but was not used in the present 11-gene analysis, which relied on explicit gene-specific priors.

Candidate cutoffs were then scanned deterministically within the central tumor-expression range defined by the minimum group proportion parameter minprop m (default $m = 0.25$). Let $Q_p(x)$ denote the empirical p -th quantile of the winsorized tumor-expression vector x . The admissible search interval was therefore

$$(Q_m(x), Q_{1-m}(x)),$$

with minprop = 0.25 in the current runner.

With the current runner, candidate cutoffs were generated on a quantile grid within this interval. When grid = "quantile", candidate cutoffs were generated as unique quantile-grid values inside this admissible interval rather than by continuous optimization. Accordingly, candidate cutoffs were evaluated only between the 25th and 75th percentiles of the tumor-expression distribution. Rather than performing continuous optimization over this interval, the primary scan assessed a fixed set of quantile-based candidate values, generated at a resolution corresponding to the default grid density of 200 and then collapsed to unique admissible cutoffs within the allowed range.

For each candidate cutoff c , samples were dichotomized into

$$gi(c) = \begin{cases} \text{High, } xi > c, \\ \text{Low, } xi \leq c. \end{cases}$$

A candidate was discarded if it violated any feasibility criterion: minimum group proportion $< m$, fewer than 5 events in either group, failed Cox fitting, non-finite coefficients, non-finite standard errors, $se > 3$, or $|\beta_c| > 5$, where β_c is the dichotomized Cox coefficient. These latter filters were implemented as practical guards against unstable edge-peak or separation-like solutions. If `expected_dir` was not NULL, an additional direction constraint was imposed: $HR(c) < 1$ was required for `protective_high`, and $HR(c) > 1$ was required for `adverse_high`.

Among the remaining valid candidates, the selected cutoff c^* was chosen by a fixed lexicographic ordering rule. Under the primary cutoff-selection rule used in the present analysis (stored internally as `max_wald_then_min_p`), candidates were ordered by decreasing score, then increasing Wald-test P value, and finally by increasing $|HR(c) - 1|$ as a final deterministic tie-breaking criterion. Under the current primary runner, score equals the Wald statistic because $\lambda = 0$ and no reference-cutoff penalty is applied. Thus, candidate cutoffs were prioritized by the largest Wald statistic, followed by the smallest Wald-test P value and, if still tied, the smallest absolute deviation of the hazard ratio from 1. This rule makes cutoff selection deterministic for a fixed dataset and decision rule rather than dependent on arbitrary tie resolution.

After selection of c^* , the final dichotomized survival association was estimated on the full per-gene tumor dataset D using $Surv(time, event) \sim I(expr > c^*)$, from which the hazard ratio $HR^* = \exp(\beta^*)$, Wald-test P value, and 95% confidence interval were extracted. A log-rank test at the same cutoff was also computed. Across genes, both the cutoff-based Cox P values and the log-rank P values were adjusted by the Benjamini–Hochberg method in the current runner.

To assess the internal stability of the selected cutoff under bootstrap resampling, StaBiCut performed bootstrap resampling with replacement and

rescanned candidate cutoffs only within a local neighborhood around c^* rather than re-optimizing over the entire expression range.

When local restriction was enabled, bootstrap resampling was designed as a local reproducibility test for the originally selected cutoff within its immediate expression neighborhood, rather than as repeated de novo optimization over the full expression range. The local search window was centered on the selected cutoff and scaled to the tumor-expression interquartile range. Specifically, if IQR_x denotes the interquartile range of the winsorized tumor-expression vector in the original dataset, the bootstrap candidate window was defined as

$$[c^* - \omega IQR_x, c^* + \omega IQR_x],$$

where ω was set to 0.5, corresponding to half of the interquartile range in the current implementation.

Accordingly, candidate cutoffs within each bootstrap replicate $D^{(b)}$ were proposed only from this local search interval, using a quantile-based grid of lower density than that used in the primary scan (bootstrap grid density 120; primary grid density 200; minprop = 0.25; minimum 5 events per group). Once a bootstrap-selected cutoff $\tilde{c}_b$ had been proposed, however, its hazard direction was re-evaluated on the full bootstrap sample before retention. Thus, local restriction constrained only the proposal of candidate cutoffs, whereas direction-consistent support still had to be demonstrated at the level of the complete resampled cohort.

Let $B = \{\tilde{c}_b\}_{b=1}^{n_{valid}}$ denote the set of valid retained bootstrap cutoffs. Bootstrap variability was characterized by its median, interquartile range,

$$IQR_{boot} = Q_{0.75}(B) - Q_{0.25}(B),$$

and empirical 95% interval width,

$$CI_{boot}^{95} = Q_{0.975}(B) - Q_{0.025}(B).$$

In the current implementation, however, these bootstrap summary statistics

were considered interpretable only when at least 50 valid bootstrap cutoffs were retained; otherwise, they were treated as unavailable for downstream use.

To compare genes on a common relative scale, both the bootstrap IQR and the empirical central 95% percentile interval width were normalized by the interquartile range of the original winsorized tumor-expression vector x :

$$IQR_{rel} = \frac{IQR_{boot}}{IQR_x}, \quad CI_{rel} = \frac{CI_{boot}^{95}}{IQR_x}.$$

The bootstrap stability component was then defined as

$$S_{boot} = \exp(-k_1 IQR_{rel}) \exp(-k_2 CI_{rel}),$$

with $k_1 = 3$ and $k_2 = 2$ in the current implementation. Narrower bootstrap dispersion relative to the intrinsic expression scale therefore yields a larger stability score. If either the bootstrap summary or IQR_x was unavailable or non-finite, this component was returned as missing.

Post-selection cutoff-position metrics were then computed. In the current implementation, this cutoff-plausibility score is stored internally as `density_score` for historical code continuity. First, cutoff centrality within the tumor-expression distribution was quantified by the empirical cumulative distribution function

$$q^* = F_x(c^*) = \frac{1}{n} \sum_{i=1}^n \mathbf{I}(x_i \leq c^*).$$

The corresponding centrality score was defined as

$$S_{plausibility} = 1 - |q^* - 0.5|.$$

This quantity is highest for cutoffs near the distributional center and lower for cutoffs closer to the tails. Although retained in the code as `density_score`, this ECDF-based metric measures distributional centrality rather than kernel density.

Second, group balance was quantified from the final dichotomization at c^* . If n_{low} and n_{high} denote the sizes of the Low and High groups, respectively, then

$$p_{min} = \frac{\min(n_{low}, n_{high})}{n}.$$

The balance component used in the composite score was

$$S_{balance} = \min\{1, \max(0, 2p_{min})\}$$

Because candidate generation had already enforced the minimum group proportion constraint, this term acts as a post-selection soft preference among admissible cutoffs rather than as the primary feasibility filter itself.

Third, tumor-normal directional concordance was evaluated by comparing the sign of the tumor-normal shift with the sign of the cutoff-based hazard direction:

$$C_{TN} = \text{sign}(\Delta_{TN}) \times \text{sign}\{\log(HR^*)\}.$$

Accordingly, $C_{TN} = 1$ indicates directional concordance and $C_{TN} = -1$ indicates directional discordance. If Δ_{TN} was unavailable, HR^* was unavailable or non-positive, or either sign was zero, this term was treated as missing. For composite scoring, the current runner maps this quantity to

$$S_{TN} = \begin{cases} 1, & C_{TN} = 1, \\ 0, & C_{TN} = -1, \\ NA, & \text{otherwise.} \end{cases}$$

Thus, under the current implementation, missing directional evidence propagates to this component rather than being replaced by an intermediate value.

Fourth, bootstrap hazard-direction consistency was used to assess whether the retained bootstrap-supported cutoffs preserved the same survival orientation as the reference cutoff. Let

$$d_0 = \text{sign}\{\log(HR^*)\},$$

denote the reference hazard direction on the original tumor dataset D . For each retained bootstrap cutoff $\tilde{c}_b$ in B , let

$$d_b = \text{sign}\{\log(HR(\tilde{c}_b; D))\},$$

where $HR(\tilde{c}_b; D)$ denotes the hazard ratio recomputed on the original tumor dataset D at cutoff $\tilde{c}_b$. Bootstrap hazard-direction consistency was then defined as the proportion of retained bootstrap cutoffs for which $d_b = d_0$, then

$$S_{\text{hazard}} = \frac{1}{M} \sum_{b=1}^M I(d_b = d_0),$$

where M is the number of bootstrap cutoffs yielding valid direction estimates on the original dataset D . In the current implementation, this quantity was returned as missing if fewer than 30 valid bootstrap cutoffs were retained, if the reference hazard direction could not be defined, or if fewer than 30 retained bootstrap cutoffs yielded evaluable hazard directions after re-estimation on the original dataset.

The overall composite prioritization score integrated the five post-selection components using fixed weights:

$$S_{\text{composite}} = 0.40 S_{\text{boot}} + 0.20 S_{\text{plausibility}} + 0.15 S_{\text{hazard}} + 0.15 S_{\text{TN}} + 0.10 S_{\text{balance}}.$$

By construction, the bootstrap stability component receives the largest weight (0.40) in the current implementation. Genes were ranked by descending $S_{\text{composite}}$, with minimum-rank assignment for ties. In the current runner, the composite score is computed directly from these five components, with no explicit handling of missing component values; accordingly, missing values in any required component can lead to an unavailable composite score.

Taken together, StaBiCut should be interpreted as a prior-aware, direction-constrained framework for prioritizing survival cutoffs, rather than as a purely data-driven search for the most statistically significant threshold. Its main mathematical contribution lies not in proposing a new survival test, but in formally integrating deterministic candidate filtering, local bootstrap support, direction-aware plausibility assessment, and fixed-weight post-selection ranking into a coherent prioritization strategy. Under this design, its selected cutoffs are intended to provide robust working thresholds for cohort-level stratification under the stated decision rules, not as exact biological boundaries.